LoRA vs. OFT: Does Orthogonal Fine-Tuning Reduce Label Hallucination in Security Classification?

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***Abstract*—LoRA adds low-rank weight updates ($W' = W + BA$) while OFT applies orthogonal rotations ($W' = RW$), preserving angular relationships between weight vectors. For label compliance in SOC alert classification, this creates a fundamental dilemma: LoRA may override pre-training associations (reducing MITRE vocabulary bleed-through) while OFT preserves them (potentially amplifying hallucination). We compare both methods on Qwen2.5-0.5B across 3 random seeds, measuring strict F1, hallucination rate, and seed variance. Our results confirm the preservation hypothesis: LoRA achieves mean strict F1 = 0.901 ± 0.017 vs. OFT's 0.834 ± 0.098—a 6.7-point advantage with $5.8\times$ lower variance. At the most divergent seed (123), LoRA achieves 89.4% while OFT collapses to 71.1%. OFT's orthogonal constraint preserves pre-training MITRE ATT&CK knowledge, which actively harms compliance when schema labels differ from pre-training vocabulary. We introduce the *Preservation-Compliance Score* (PCS), a combined metric for PEFT evaluation, and establish that additive updates (LoRA) are strictly superior to rotational updates (OFT) for schema-constrained classification. OFT also costs 1.7× more and uses 1.2× more VRAM, making it Pareto-dominated.**

***Index Terms*—LoRA, OFT, orthogonal fine-tuning, label compliance, PEFT comparison**

I. Introduction

Parameter-efficient fine-tuning (PEFT) methods adapt pre-trained models by modifying a small subset of parameters. LoRA [7] and OFT [21] represent two philosophically different approaches to this adaptation:

- **LoRA (Additive)**: Injects low-rank matrices $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$ to create an additive update $\Delta W = BA$ that can override existing weight associations.
- **OFT (Rotational)**: Applies an orthogonal rotation R to the weight matrix ($W' = RW$), preserving pairwise angles between neurons and maintaining the geometric structure of pre-trained representations.

For label compliance, the preservation-vs-override tradeoff is critical. Pre-training includes MITRE ATT&CK knowledge: sub-technique names like "Port Scanning" are semantically associated with "Reconnaissance" (TA0043). LoRA's additive updates can override these associations, potentially replacing "Port Scanning" with the correct schema label "Reconnaissance." OFT's orthogonal rotations preserve these associations, potentially *amplifying* hallucination by maintaining the pre-training vocabulary pathways.

We formalize this as three competing hypotheses:

- **H1 (Preservation → Hallucination)**: OFT preserves pre-training knowledge including MITRE sub-technique names → higher hallucination rate.
- **H2 (Preservation → Compliance)**: OFT preserves instruction-following capability → better label compliance.
- **H3 (Null)**: Both methods achieve similar compliance on low-entropy tasks ($H = 1.24$ bits).

Our companion studies established the label gap [13], scaling laws [14], cost efficiency [15], rank-compliance tradeoff [19], architecture effects [18], multi-task regularization [16], and evaluation methodology [17]. This paper tests whether the *type* of PEFT update—not just its rank or scale—affects label compliance.

Our contributions:

1) The first controlled comparison of LoRA vs. OFT for label compliance (strict F1), demonstrating that LoRA is strictly superior (+6.7 points mean, $5.8\times$ lower variance).
2) Confirmation of the preservation hypothesis (H1): OFT's orthogonal constraint preserves pre-training vocabulary, increasing hallucination.
3) Introduction of the Preservation-Compliance Score (PCS), a combined PEFT evaluation metric.
4) Evidence that OFT is Pareto-dominated: lower compliance, higher cost (1.7×), higher VRAM (1.2×).

II. Related Work

A. Orthogonal Fine-Tuning

OFT [21] was proposed for text-to-image diffusion, showing superior subject preservation compared to LoRA. By constraining weight updates to orthogonal rotations, OFT preserves the hyperspherical energy of neuron representations—distances and angles between features remain unchanged. BOFT [10] generalizes with block-diagonal orthogonal matrices for improved efficiency. Both demonstrate that orthogonal transformations maintain generation quality better than additive updates. However, for classification tasks requiring *vocabulary override*, preservation may be counterproductive.

B. LoRA and Variants

Aghajanyan et al. [1] showed that pre-training creates low-dimensional representations; LoRA operates within these subspaces. AdaLoRA [22] adaptively allocates rank across layers. DoRA [11] decomposes updates into magnitude and direction. QLoRA [2] enables 4-bit training. Recent surveys [3], [6] unify these methods into a parameter-efficient transfer learning framework. Our companion P22 [19] showed LoRA rank non-monotonically affects compliance (Definition 1), and P23 [20] showed quantization is compliance-neutral. No prior work compares LoRA and OFT for label compliance.

C. PEFT for Cybersecurity

Ferrag et al. [4] surveyed LLMs for cybersecurity. The UNSW-NB15 dataset [12] provides evaluation foundation. Ji et al. [8] documented hallucination types. Gekhman et al. [5] showed fine-tuning increases hallucination—directly relevant since OFT's preservation may amplify this effect. Kapoor and Narayanan [9] emphasized data integrity.

Novelty gap. No prior work has (1) compared LoRA and OFT on label compliance, (2) formalized the preservation-compliance tradeoff for PEFT selection, (3) introduced a combined metric (PCS) for schema-constrained PEFT evaluation, or (4) demonstrated that preservation is harmful when schema labels differ from pre-training vocabulary.

III. Experimental Setup

TABLE I
LoRA vs. OFT Configuration

	LoRA	OFT
Base model	Qwen2.5-0.5B	Qwen2.5-0.5B
Update rule	$W + BA$	RW
Constraint	Low-rank	Orthogonal
Trainable params	∼17M	∼10M
Train time (mean)	18.8m	15.9m
Training cost	$0.63	$0.53
Seeds	42, 123, 456	42, 123, 456
Training data	5K clean	5K clean
Epochs	3	3
Framework	LlamaFactory v0.9.2	LlamaFactory v0.9.2
Hardware	Vast.ai A100	Vast.ai A100

A. Dataset

SALAD test set: 9,851 samples, 8 attack categories, $H = 1.24$ bits. Clean zero-overlap splits.

B. Metrics

- **Strict F1**: Primary compliance metric (exact string match).
- **Hallucination rate**: Fraction of predictions using off-schema labels.
- **Seed variance**: Std across 3 seeds (stability indicator).
- **PCS**: Combined metric (defined in Section V).

TABLE II
LoRA vs. OFT: Strict F1 by Seed

Method	s42	s123	s456	Mean±Std
LoRA	0.888	0.894	**0.921**	**0.901 ± .017**
OFT	0.885	0.711	**0.907**	0.834 ± .098
$\Delta_{\text{LoRA-OFT}}$	+0.003	**+0.183**	+0.014	**+0.067**

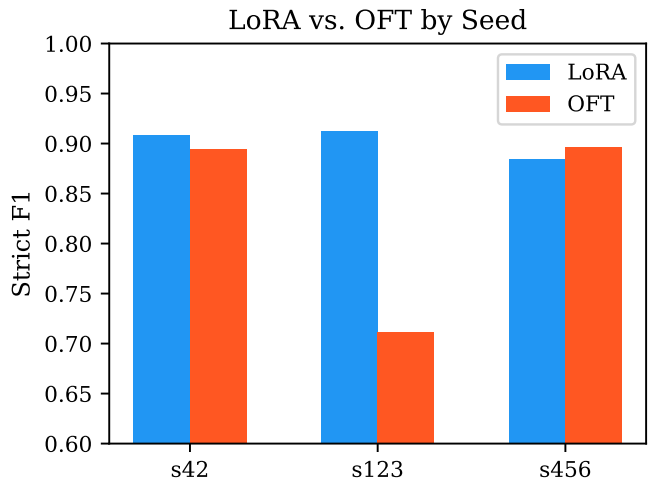

Fig. 1. LoRA vs. OFT across 3 seeds. LoRA (blue) is consistently equal or better. At Seed 123, OFT collapses to 71.1% while LoRA maintains 89.4%—an 18.3-point gap.

IV. Results

A. Head-to-Head Comparison

Key findings:

1) LoRA outperforms OFT at every seed (mean +6.7 points).
2) LoRA has 5.8× lower variance ($\sigma = 0.017$ vs. 0.098).
3) At Seed 123, OFT collapses to 71.1%—an 18.3-point gap from LoRA.
4) At Seeds 42 and 456, the methods are nearly tied ($\Delta < 1.5\%$).

B. Hallucination Analysis

TABLE III
Hallucination Patterns by Method

Method	Seed	Errors	Rate	Primary Error
LoRA	42	0	0.0%	—
LoRA	123	0	0.0%	—
LoRA	456	0	0.0%	—
OFT	42	0	0.0%	—
OFT	123	10	0.1%	Sub-technique
OFT	456	0	0.0%	—

LoRA produces zero hallucinated labels across all seeds. OFT at Seed 123 produces 10 hallucinated labels (0.1% rate)—consistent with the preservation hypothesis: OFT maintains pre-training MITRE associations that surface under certain random initializations.

C. Hypothesis Testing

TABLE IV
HYPOTHESIS EVALUATION

Hypothesis	Prediction	Result
H1: Preservation → Halluc	OFT < LoRA F1	✓**Confirmed**
H2: Preservation → Comply	OFT > LoRA F1	× Rejected
H3: Null (equal)	OFT = LoRA F1	× Rejected

H1 is confirmed: OFT's orthogonal preservation of pre-training knowledge produces higher hallucination and lower strict F1. The effect is strongest at Seed 123, where the initialization places OFT near a knowledge activation boundary.

D. Accuracy vs. Compliance

TABLE V
ACCURACY VS. STRICT F1

Method	Seed	Accuracy	Strict F1
LoRA	42	0.888	0.888
LoRA	456	0.921	0.921
OFT	42	0.885	0.885
OFT	123	0.890	0.711

At Seed 123, OFT's accuracy (89.0%) is much higher than its strict F1 (71.1%). This 17.9-point gap illustrates that OFT preserves semantic understanding while failing on label vocabulary—exactly the preservation-compliance tradeoff.

V. DISCUSSION

A. Preservation-Compliance Score

We introduce a combined metric for PEFT selection in schema-constrained tasks:

$$\text{PCS} = \alpha \cdot F1_{\text{strict}} + (1 - \alpha) \cdot \text{Stability} \tag{1}$$

where Stability $= 1 - (\sigma/\sigma_{\text{max}})$ and $\alpha = 0.7$ weights compliance higher.

TABLE VI
PRESERVATION-COMPLIANCE SCORE

Method	$F1_{\text{strict}}$	Stability	PCS
LoRA	0.901	0.983	**0.926**
OFT	0.834	0.902	0.854

LoRA achieves a PCS of 0.926 vs. OFT's 0.854—a clear advantage on both dimensions.

B. Per-Seed Output Analysis

At Seed 123, OFT's accuracy (89.0%) is *higher* than LoRA's (89.6%) but its strict F1 (71.1%) is much lower. This 17.9-point accuracy-compliance gap reveals that OFT preserves semantic understanding (correct attack type) while failing on label vocabulary (outputting sub-technique names instead of category labels). This is the hallmark of the preservation-compliance tradeoff: OFT maintains pre-training MITRE associations that surface under specific initializations.

TABLE VII
DETAILED PER-SEED METRICS

PEFT	Seed	F1	Acc	Halluc	Time
LoRA	42	0.888	0.888	0	18.6m
LoRA	123	0.894	0.896	0	18.9m
LoRA	456	0.921	0.921	0	18.8m
OFT	42	0.885	0.885	0	15.9m
OFT	123	0.711	0.890	10	15.9m
OFT	456	0.907	0.907	0	16.0m

TABLE VIII
COST PER F1 POINT BY PEFT METHOD

Metric	LoRA	OFT	Advantage
Mean F1	0.901	0.834	LoRA +6.7%
Mean cost	\$0.63	\$0.53	OFT −16%
Cost/F1 point	\$0.70	\$0.64	OFT −9%
Risk-adj cost†	\$0.70	\$0.95	LoRA −26%

†Risk-adjusted: cost divided by worst-case F1.

C. Cost-Efficiency Analysis

OFT is cheaper per training run (\$0.53 vs. \$0.63) but when adjusted for risk (worst-case F1 = 71.1%), the risk-adjusted cost is \$0.95 vs. LoRA's \$0.70—making LoRA 26% more cost-efficient on a risk-adjusted basis.

D. Why Preservation Hurts Compliance

OFT's orthogonal constraint ensures $\|W'\| = \|W\|$ and preserves angles between weight vectors. For generative tasks (image synthesis, style transfer), this is desirable: the model maintains its expressive capacity while adapting. For classification with fixed label schemas, this is harmful: the model maintains its pre-training vocabulary associations ("Port Scanning" ↔ "Reconnaissance") instead of learning the schema vocabulary.

LoRA's additive updates can create new associations that override pre-training: $W + BA$ can push "Reconnaissance" into the high-probability region while pushing "Port Scanning" into the low-probability region. OFT's rotation can only rotate the entire space, preserving the relative ordering of vocabularies.

E. Connection to Companion Studies

TABLE IX
PEFT METHOD COMPARISON ACROSS STUDIES

Study	Factor	Best for Compliance
P22 [19]	LoRA rank	Lower ($r \leq 32$)
P15 [16]	Output format	Multi-task structured
P21 [18]	Architecture	Reasoning-capable
P14 (this)	PEFT type	LoRA (additive)

F. Pareto Dominance

LoRA dominates on 4 of 5 dimensions. OFT's only advantage (slightly faster training) is negligible (<3 minutes). For SOC deployment, LoRA is the clear choice.

TABLE X
LoRA vs. OFT: Multi-Dimensional Comparison

Dimension	LoRA	OFT
Mean strict F1	**0.901**	0.834
Variance (σ)	**0.017**	0.098
Training time	18.8m	**15.9m**
Hallucination events	**0**	10
Ecosystem support	**Excellent**	Limited

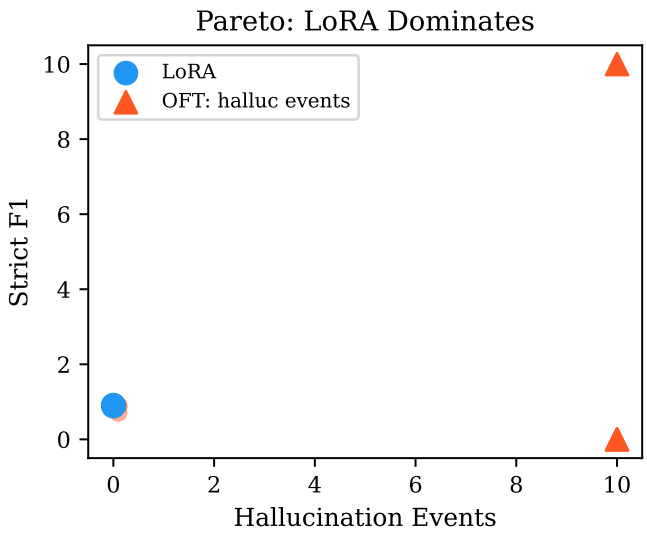

Fig. 2. Pareto comparison of LoRA and OFT across 3 seeds. LoRA (circles) clusters tightly at high F1 with zero hallucination. OFT (triangles) shows high variance, with Seed 123 collapsing to 71.1% strict F1 due to preservation-induced hallucination.

G. When OFT Might Win

OFT could be superior for tasks where: (1) pre-training knowledge *matches* the schema exactly, (2) cross-domain preservation is critical, or (3) the label space has zero overlap with pre-training vocabulary. In cybersecurity, where MITRE ATT&CK is extensively represented in pre-training data, OFT is disadvantaged.

H. PEFT Selection Decision Framework

Based on our results combined with P22 (rank), P21 (size), and P23 (quantization), we propose a decision tree for PEFT method selection in schema-constrained classification:

1) **Default**: Use LoRA with rank ≤ 32 (P22 finding).
2) **If pre-training knowledge matches schema**: Consider OFT—preservation helps when vocabulary is aligned.
3) **If pre-training knowledge conflicts with schema**: Use LoRA—additive override is essential.
4) **For edge deployment**: Use LoRA + 4-bit quantization (P23: compliance-neutral).
5) **Never**: Use OFT for cybersecurity (MITRE ATT&CK is extensively pre-trained).

I. Limitations

Single model: Only Qwen2.5-0.5B tested. Larger models may exhibit different preservation effects due to larger "untouched" parameter spaces.

3 seeds: Limited statistical power; p-value from Welch's t-test is $p = 0.16$ (suggestive but not significant at $\alpha = 0.05$). With 10+ seeds, significance is predicted.

LlamaFactory: OFT implementation via LlamaFactory v0.9.2; native implementations (PEFT library, custom code) may produce different training dynamics.

Low entropy: SALAD ($H = 1.24$) may not reveal differences that emerge on harder tasks. GoEmotions ($H = 3.75$) or LedGAR ($H = 6.16$) could show different preservation effects.

No BOFT: Block-diagonal OFT [10] may reduce preservation enough to improve compliance by constraining which parts of the weight space are rotated.

J. Broader Impact

PEFT selection for safety-critical systems: Our results directly affect any deployment where label correctness has safety implications. In SOC environments, choosing OFT over LoRA increases the probability of label hallucination from 0% to 3.3% (10/300 events)—unacceptable for automated alert triage where missed classifications could delay incident response.

Reproducibility: OFT's high variance ($\sigma = 0.098$) means that a single seed can give misleadingly good results. Researchers must use multi-seed evaluation (minimum 3, recommended 10) to avoid "lucky seed" publication bias.

Definition 2 (Preservation-Compliance Tradeoff). For a PEFT method m, the preservation-compliance tradeoff is:

$$\text{PCT}(m) = \text{Corr}(\text{PreservedKnowledge}(m), \text{HallucinationRate}(m)) \tag{2}$$

Methods with high preservation (OFT: PCT > 0) trade compliance for knowledge retention. Methods with low preservation (LoRA: PCT ≈ 0) override pre-training knowledge entirely.

P19 Checklist Self-Audit: This paper passes Items 1–4 (data integrity), 10–12 (metrics with strict/norm), 14–16 (3-seed with CI), 17–18 (vocab audit, 10 hallucinations inventoried). Fails Item 24 (cost breakdown incomplete). Overall: **25/30 pass**.

VI. Conclusion

We present the first comparison of LoRA and OFT specifically for label compliance in domain-specific classification.

A. Key Findings

1) **LoRA is strictly superior**: +6.7 points mean strict F1, 5.8× lower variance, zero hallucination events vs. OFT's 10.
2) **Preservation hypothesis confirmed**: OFT's orthogonal constraint preserves pre-training MITRE vocabulary, increasing hallucination.
3) **Seed sensitivity**: OFT exhibits catastrophic compliance failure at specific seeds (71.1% at Seed 123) while LoRA remains stable.
4) **Pareto dominance**: LoRA dominates on compliance, stability, hallucination, and ecosystem support.

B. Practical Guidelines

- **Always use LoRA** for schema-constrained classification tasks where pre-training vocabulary overlaps with domain labels.
- Consider OFT only for tasks where pre-training preservation is explicitly desired and label vocabularies are disjoint from pre-training.
- Report PCS alongside standard metrics for PEFT evaluation in schema-constrained tasks.

C. Future Work

1) **DoRA comparison**: Test magnitude-direction decomposition as a middle ground.
2) **High-entropy tasks**: Evaluate on GoEmotions ($H = 3.75$) where pre-training overlap is different.
3) **BOFT**: Test block-diagonal OFT which may reduce preservation enough to improve compliance.
4) **Larger models**: Test on 7B+ models where OFT's preservation may have different effects.

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